Positive Square Energy of Every Tetracyclic Graph

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Abstract

Let G be a finite simple connected graph with n vertices and $n+3$ edges. We prove that the sum $s^+(G)$ of the squares of its positive adjacency eigenvalues is at least n. The positive cyclomatic ranks of the cyclic blocks partition four as $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, or 4. The first four alternatives are closed by a sharp block-additive doubly nonnegative bound and explicit induced-territory packets for its residuals. In the last alternative, suppression of degree-two paths gives exactly seventeen loopless 2-connected multigraph kernels. We include the DNN correlation dual, exact path elimination, the kernel classification, and complete count ledgers for the exact finite certificates covering all seventeen kernels. Parity monotonicity turns each finite frontier into an all-length proof, and one-vertex additivity includes arbitrary rooted trees. Some subsidiary estimates are strict, but boundary certificates occur, so no strict global assertion is made.

1 Statement and spectral reductions

All graphs are finite, simple, and undirected. Let $\lambda_1, \dots, \lambda_n$ be the adjacency eigenvalues and put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \qquad D(G) = s^+(G) - s^-(G).$$

Since $\operatorname{tr} A(G)^2 = 2|E(G)|$, a tetracyclic graph satisfies

$$s^+(G) + s^-(G) = 2n + 6, \qquad s^+(G) = n + 3 + \frac{1}{2} D(G). \tag{1}$$

Positive and negative square energies were introduced in connection with spectral coloring bounds [1]. The general lower bound $s^+(G) \geq n-1$ is due to Liu–Tang–Zhang [3]. The present result settles the $m = n+3$ connected frontier of the edge-surplus conjecture of Akbari–Kumar–Mohar–Pragada–Zhang [2]; it does not address graphs with more edges by an edge-addition argument.

Theorem 1.1. *Every finite simple connected graph G with $|V(G)| = n$ and $|E(G)| = n+3$ satisfies*

$$\boxed{s^+(G) \geq n}.$$

We use square-energy credit $\sigma(H) = s^+(H) - |V(H)|$. The following standard pinching inequality permits structural repairs.

Lemma 1.2 (Induced superadditivity)**.** *If $V(G) = V_1 \dot\cup \cdots \dot\cup V_r$, then*

$$s^+(G) \geq \sum_{j=1}^{r} s^+(G[V_j]), \qquad \sigma(G) \geq \sum_{j=1}^{r} \sigma(G[V_j]).$$

Proof. For a real symmetric matrix A,

$$\operatorname{tr}(A_+^2) = \max_{X \succeq 0} \bigl(2\operatorname{tr}(AX) - \operatorname{tr}(X^2)\bigr).$$

Insert the block diagonal matrix whose blocks are the positive parts of the principal matrices $A[V_j]$. Cross edges make no contribution to the trace. □

A nonempty connected tree T is bipartite, so $s^+(T) = |E(T)| = |V(T)| - 1$ and $\sigma(T) = -1$. We shall use the following attachment-uniform credits, all proved in the cited lower-rank manuscripts.

Lemma 1.3 (Lower-rank packets)**.** *Arbitrary rooted trees may be attached at arbitrary vertices in every item.*

1. *Every connected bicyclic graph has positive credit; a bipartite one has credit exactly one. A two-triangle cactus and $\Theta(1,2,2)$ have credit strictly greater than one.*
2. *Every connected tricyclic graph has nonnegative credit.*
3. *A triangular unicyclic cactus has positive credit. A pentagonal unicyclic cactus has credit at least $2 - \sqrt{5}$.*
4. *A triangle–pentagon bicyclic cactus has credit greater than $3 - \sqrt{5}$, and a three-cycle packet C_3, C_3, C_5 whose triangles share a cut has credit greater than $4 - \sqrt{5}$.*

Proof. The bicyclic and tricyclic statements are the complete lower-rank theorems [4, 5]. The sharper cactus credits follow from the signed Coulson formula after rooted-tree matching-message elimination [6]. They are quoted with exactly the margins used below; no edge-addition monotonicity is involved. □

2 The DNN dual and exact paths

For an edge-containing graph H, define

$$\kappa(H) = \sup_{\substack{M \succeq 0,\ M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4\bigl(\sum_{uv \in E(H)} \sqrt{M_{uv}}\bigr)^2}{\mathbf{1}^T M \mathbf{1}}. \tag{2}$$

Lemma 2.1 (DNN inequality and correlation dual)**.** *For every graph H,*

$$s^-(H) \leq \kappa(H), \qquad \kappa(H) = \min_{\substack{R \succeq 0 \\ R_{vv} = 1}} \sum_{uv \in E(H)} \frac{2}{1 - R_{uv}}, \tag{3}$$

with boundary values interpreted by limits. If $H = H_1 \vee H_2$ is a one-vertex sum, then

$$\kappa(H) = \kappa(H_1) + \kappa(H_2).$$

In particular, $\kappa(T) = |E(T)|$ for every tree.

Proof. Write $A = P - B$ for the positive and negative spectral parts. The Schur product theorem makes $M = B \circ B$ positive semidefinite and entrywise nonnegative. Moreover

$$\mathbf{1}^T M \mathbf{1} = \operatorname{tr} B^2 = s^-(H), \qquad \frac{s^-(H)}{2} = -\sum_{uv \in E(H)} B_{uv} \leq \sum_{uv \in E(H)} \sqrt{M_{uv}}.$$

Substitution in (2) proves $s^- \le \kappa$.

For weak duality, if R is a correlation matrix, then

$$\mathbf{1}^T M \mathbf{1} - \sum_{uv \in E(H)} 2(1 - R_{uv}) M_{uv} = \langle R, M \rangle + 2 \sum_{\substack{u<v \\ uv \notin E(H)}} (1 - R_{uv}) M_{uv} \ge 0.$$

Weighted Cauchy–Schwarz gives the upper bound in (3). For the reverse bound normalize $\langle J, M \rangle = 1$, use

$$(\sum_e \sqrt{x_e})^2 = \min_{a_e > 0,\ \sum a_e = 1} \sum_e \frac{x_e}{a_e},$$

and apply conic minimax with $(\mathcal{S}_+ \cap \mathcal{N})^* = \mathcal{S}_+ + \mathcal{N}$; an ε truncation treats boundary points.

For a one-vertex sum, restriction of a feasible correlation matrix gives one inequality. Conversely, realize the two correlation matrices by unit Gram vectors with their common cut vector equal and place their orthogonal components in orthogonal subspaces. This glues feasible matrices and proves the opposite inequality. Iterating $\kappa(K_2) = 1$ proves the tree formula. □

Lemma 2.2 (Exact path elimination)**.** *Replace a kernel edge uv by a path of length ℓ. For fixed endpoint correlation $r = R_{uv}$, the exact excess of its contribution over ℓ is*

$$f_\ell(r) = \ell \tan^2\left(\frac{\arccos((-1)^\ell r)}{2\ell}\right). \tag{4}$$

For fixed r and fixed parity, $f_{\ell+2}(r) \le f_\ell(r)$, strictly unless the transformed endpoint angle is zero. If a subdivided kernel has total path length L, then

$$\kappa(H) - L = \min_R \sum_P f_{|P|}(R_{u_P v_P}), \tag{5}$$

where R ranges over correlation matrices on the branch vertices.

Proof. Represent R by unit vectors and alternately negate the vectors along the path. The transformed endpoint angle is $\beta = \arccos((-1)^\ell r)$. The spherical triangle inequality makes the sum of successive angles at least β, and convexity of $\sec^2(x/2)$ makes equal angles optimal. Equally spaced planar vectors on a shortest arc attain equality, yielding $\ell \sec^2(\beta/(2\ell)) = \ell + f_\ell(r)$. The constructions for distinct paths share only their fixed endpoint vectors and are therefore compatible. Finally, for $h_s(x) = s \tan^2(x/s)$,

$$\frac{d}{ds} h_s(x) = \tan z \sec^2 z \big(\sin z \cos z - 2z\big) < 0, \qquad z = x/s,$$

because $0 < \sin z \cos z < z < 2z$. This proves fixed-parity monotonicity. □

If a connected rank-r core has L edges, it has $L - r + 1$ vertices. Consequently, for rank four,

$$\kappa(H) \le L + 3 \quad \Longrightarrow \quad s^+(H) \ge 2L - (L+3) = L - 3 = |V(H)|. \tag{6}$$

Tree attachments preserve this calculation, since each new tree edge adds one to L, $|V|$, and κ.

3 Block partitions and all multiblock residuals

A block is a maximal connected subgraph without a cut vertex; bridges count as two-vertex blocks. Cyclomatic rank is additive over blocks. Suppression and all degree statements below are internal to the relevant block.

Proposition 3.1 (Block alternatives)**.** *The positive ranks of the cyclic blocks of a connected tetracyclic graph are exactly*

$$1+1+1+1, \qquad 2+1+1, \qquad 2+2, \qquad 3+1, \qquad 4.$$

Rank-one blocks are cycles, and rank-two blocks are subdivisions of three parallel kernel edges, i.e. theta graphs.

Proof. The positive block ranks are positive integers summing to four. A rank-one 2-connected block is a cycle. After suppressing degree-two vertices of a rank-two block, the identity $\sum_v (d(v)-2) = 2$ and 2-connectivity force two degree-three vertices joined by three parallel edges. Suppression creates no loop: a suppressed loop would be a cycle attached to the rest at one branch vertex, making that vertex a cut. □

For a theta with path lengths a, b, c, define its exact DNN excess

$$\Delta(a,b,c) = \min_{0\le\theta\le\pi} \left(\sum_{\ell \text{ even}} \ell \tan^2 \frac{\theta}{2\ell} + \sum_{\ell \text{ odd}} \ell \tan^2 \frac{\pi-\theta}{2\ell} \right). \tag{7}$$

For a cycle let $\epsilon_q = 0$ for even q and $\epsilon_q = q\tan^2(\pi/(2q))$ for odd q. Lemma 2.2 gives $\kappa(\Theta) = |E(\Theta)| + \Delta$ and $\kappa(C_q) = q + \epsilon_q$.

The elementary tangent classification needed here is

$$\Delta(a,b,c) > 1 \iff (a,b,c) = (1,2,r)\ (r \ge 2), \qquad \max \Delta = \Delta(1,2,2) = \frac{\sqrt{17}-1}{2}, \tag{8}$$

and, away from $(1,2,2)$ in the exceptional family, $\Delta(1,2,r) < 4/3$. Also

$$\epsilon_3 = 1, \qquad \epsilon_5 = 5 - 2\sqrt{5} < \frac{2}{3}, \qquad \epsilon_q < \frac{2}{5} \quad (q \ge 7 \text{ odd}). \tag{9}$$

Here is the full classification argument. Put $x = \theta/2$ and $h_s(y) = s\tan^2(y/s)$. For fixed $0 < y < \pi/2$,

$$\frac{d}{ds} h_s(y) = \frac{\tan(y/s)}{\cos^2(y/s)} \left(\sin(y/s)\cos(y/s) - \frac{2y}{s} \right) < 0. \tag{10}$$

Thus an upper-bound test may replace an even path by length two and a nonunit odd path by length three. Simplicity permits at most one unit path.

Suppose first that there is no unit path, and let e be the number of even paths. It is enough to test

$$F_e(x) = 2e\tan^2(x/2) + 3(3-e)\tan^2((\pi/2 - x)/3).$$

For $e = 0, 3$, use respectively $x = \pi/2, 0$. For the mixed cases use

$$F_1(\pi/4) < 1, \qquad F_2(\pi/6) < 1.$$

The identities $\tan(\pi/8) = \sqrt{2} - 1$ and $\tan(\pi/12) = 2 - \sqrt{3}$, together with $\tan(\pi/9) < \tan(\pi/8)$, bound the two values by $48 - 4\sqrt{2} - 24\sqrt{3}$ and $37 - 6\sqrt{2} - 16\sqrt{3}$; successive squaring of positive sides proves both are below one. Hence $\Delta < 1$ in this case.

Now suppose one path has length one. Its contribution is $\cot^2 x$. If the other paths are both odd, $x = \pi/2$ gives zero. If exactly one is even and that even length is at least four, $x = \pi/3$ gives

$$\Delta \le \frac{1}{3} + 4\tan^2\frac{\pi}{12} + 3\tan^2\frac{\pi}{18} < 1;$$

if both are even and at least four, the same test gives

$$\Delta \le \frac{1}{3} + 8\tan^2\frac{\pi}{12} < 1.$$

Indeed $\tan(\pi/18) < \tan(\pi/12) < 1/3$, and both comparisons reduce to $48\sqrt{3} > 83$, equivalently $6912 > 6889$ after squaring. Therefore the only remaining sorted triples are $(1, 2, r)$, $r \ge 2$. With $u = \tan(x/2)$, the first two terms obey

$$\cot^2 x + 2\tan^2(x/2) = \frac{1}{4u^2} - \frac{1}{2} + \frac{9}{4}u^2 \ge 1.$$

The third term is positive in the interior; at either endpoint where it can vanish, the displayed expression diverges or equals two. Compactness makes the minimum strictly greater than one. This proves the equivalence in (8).

For $r = 2$ the complete objective is

$$\frac{1}{4u^2} - \frac{1}{2} + \frac{17}{4}u^2,$$

whose minimum, at $u^2 = 1/\sqrt{17}$, is $(\sqrt{17}-1)/2$. For even $r \ge 4$, (10) makes the objective pointwise smaller. For odd $r \ge 3$, testing $x = \pi/3$ and using $\tan(\pi/18) < 1/3$ gives $\Delta < 4/3$; for even $r \ge 4$ the same test gives $1 + 4(2 - \sqrt{3})^2 < 4/3$. Thus $(1, 2, 2)$ is the global maximum and every other exceptional row has the asserted strict $4/3$ bound. Finally, $\epsilon_q = h_q(\pi/2)$ decreases through odd q by (10); direct evaluation gives $\epsilon_3 = 1$ and $\epsilon_5 = 5 - 2\sqrt{5}$, while $\sin t < t$, $\cos t > 1 - t^2/2$, and $\pi < 22/7$ at $t = \pi/14$ give $\epsilon_7 < 2/5$. This proves (9) and reproduces the complete theta–cycle classification from the tricyclic proof [5].

Proposition 3.2 (Multiblock closure)**.** *Every tetracyclic graph having at least two positive-rank cyclic blocks satisfies* $s^+(G) \ge |V(G)|$.

Proof. We give the complete residual ledger. In each direct row, block additivity of κ, together with (6), proves the claim. The last column names the only rows not direct and the packet that closes them.

block ranks	direct DNN criterion	exact residual packet
1^4	$\sum_i \epsilon_{q_i} \le 3$	$\{3, 3, 3, q\}$ with odd q, and $\{3, 3, 5, 5\}$; the complete cactus territory theorem closes both families
$2 + 1 + 1$	$\Delta + \epsilon_p + \epsilon_q \le 3$	$\Theta(1, 2, r) + C_3 + C_3$ and $\Theta(1, 2, 2) + C_3 + C_5$
$2 + 2$	$\Delta_1 + \Delta_2 \le 3$	two copies of $\Theta(1, 2, 2)$
$3 + 1$	$e_B + \epsilon_q \le 3$ for each rank-three certificate	the canonical doubled-triangle and doubled-C_4 class 111 rows, and the all-odd K_4 rows with at most one long path

For 1^4, the exact cycle calculation leaves precisely the two displayed families: $\epsilon_3 = 1$, and among two nontriangular odd cycles only the pair $5, 5$ has excess sum greater than one. Maximum-packing territories, shared-cut incidence trees, and the exact twenty-type C_3, C_3, C_5, C_5 ledger close them, including arbitrary connectors and rooted branches [6].

For $2+1+1$, equations (8)–(9) leave exactly the two rows shown. In $\Theta(1,2,r)+C_3+C_3$, write the theta paths as the edge xy, the length-two path xay, and the x–y path P of length r. Open an internal vertex v of P. Its territory R consists of v, every off-theta component whose block-cut route first meets the theta at v, and every rooted branch based there. The complementary induced territory H is connected through the other two theta arms and contains the intrinsic triangle $xayx$; the remnants of $P - v$ are trees in H. Every cut vertex belongs to exactly one side, and every component at that cut follows its unique block-cut-tree route.

Let k be the number of external triangles owned by R. For $k = 1$, R is a triangular packet of credit > 0 and H is a two-triangle packet of credit > 1; for $k = 2$ these credits are reversed. Let $k = 0$. Then R is one nonempty tree of credit -1. If the three triangles retained in H have packing number at most two—in particular, if the external triangles share a cut—their normalized Sachs polynomial has strictly negative imaginary part: singleton triangle terms are negative imaginary and, since no three cycles are vertex-disjoint, all other cycle-collection terms are real. Hence $D(H) > 0$, and because H is tricyclic its credit is $2 + D(H)/2 > 2$, which pays R. If their packing number is three, the triangles are pairwise vertex-disjoint. The reduced block tree joining them contains an actual bridge; cut one such bridge so that the two retained induced cyclic territories contain two triangles and one triangle. Give the whole connector component at the cut to one endpoint territory. Their credits are > 1 and > 0, so together with R the total is strictly greater than $1 + 0 - 1 = 0$. This exhausts $k = 0, 1, 2$ and specifies ownership of every connector, cut, and rooted branch.

For $\Theta(1,2,2)+C_3+C_5$, the shared-cut incidence tree has either separate clusters or one central block. Separating bridges use the credits > 1, > 0, and $\geq 2-\sqrt{5}$. In one cluster, opening the theta internal vertex away from its triangle cut leaves either a two-triangle packet against a pentagonal packet, or a shared-triangle C_3, C_3, C_5 packet against one tree. Their credits are respectively $> 1 + (2 - \sqrt{5}) > 0$ and $> (4 - \sqrt{5}) - 1 > 0$. If the external triangle or pentagon is the central block, split that cycle at one cut; the remaining path belongs to the packet at the other cut. These are all central-node choices and assign every cut and branch once [14].

For $2 + 2$, (8) shows that only two diamonds $\Theta(1,2,2)$ can exceed the budget. If an actual bridge separates them, cut it and assign the complete connector to either endpoint territory; the two attached-diamond credits are each > 1. Suppose instead that they share a cut w, and keep w, the first diamond, and all branches assigned to that side in a territory H. If w has degree three in the second diamond, its other three vertices induce a path, which with their rooted branches is one nonempty tree T. If w has degree two there, choose a degree-three neighbor p and transfer p, the edge wp, and every branch rooted at p to H. This adds only a rooted tree at w, so H is still an attached-diamond packet; the two vertices left from the second diamond induce an edge and, with their branches, form one nonempty tree T. In both cut orbits the territories are disjoint, connected, and induced, every cut and branch is owned exactly once, and $\sigma(H) + \sigma(T) > 1 - 1 = 0$. This is the complete two-diamond argument [16].

For $3 + 1$, use the complete rank-three ledgers from the tricyclic proof [5]. The following table makes the canonical/noncanonical split exhaustive. Here j is the number of long paths in the all-odd K_4 class; "canonical pair" means lengths $\{1, 2\}$ in each doubled pair. Every number in the fourth column is an exact rational upper bound, not a decimal optimization.

rank-three family	physical state	number/disposition	actual excess certificate
four-path theta	all simple rows	all DNN	$e_B \le 2$; equality only at $(1,2,2,2)$
doubled triangle	outside class 111	28 labelled rows, DNN	$e_B \le 2$ from the physical Gram ledger
doubled triangle, class 111	a parallel path noncanonical	four relabelled choices, DNN	even-long $e_B < 229/120 < 2$; odd-long $e_B < 31/20 < 2$
doubled triangle, class 111	both doubled pairs canonical	four labelled rows, structural	hostile-cycle certificate $e_B < 221/100$
doubled C_4	outside class 111	28 labelled rows, DNN	$e_B \le 2$ from the physical Gram ledger
doubled C_4, class 111	a doubled path noncanonical	either pair and either parity, DNN	even-long $e_B < 1862/1000 < 2$; odd-long $e_B < 1662/1000 < 2$
doubled C_4, class 111	both doubled pairs canonical	eight labelled rows, structural	hostile-cycle certificate $e_B < 9/4$
K_4, non-all-odd	all physical rows	56 labelled rows, DNN	$e_B \le 5/3 < 2$
K_4, all odd	physically all odd, $j \ge 3$	all subsets, DNN	simplex: $e_B < 3 - j/3 \le 2$
K_4, all odd	switched, $j \ge 4$	three four-edge cuts, DNN	simplex: $e_B < 3 - j/4 \le 2$
K_4, all odd	switched, $j = 3$	four three-edge cuts, DNN	physical Gram: $e_B = (837 - 168\sqrt{21})/35 < 2$
K_4, all odd	$j = 2$, opposite	three placements, DNN	$e_B \le 24 - 16\sqrt{2} < 2$
K_4, all odd	$j = 2$, adjacent	twelve placements, DNN	$e_B < 71/40 < 2$
K_4, all odd	$j = 1$	six placements, structural	hostile-cycle certificate $e_B < 12/5$
K_4, all odd	$j = 0$	the unsubdivided K_4, structural	attached-K_4 Sachs packet

The two doubled-triangle bounds come from the planar angle records $(12; 16, 7)$ and $(6; 2, 8)$ at physical lengths $(4, 1, 2, 1, 1)$ and $(2, 3, 2, 1, 1)$. The doubled-C_4 bounds come from $Q(0, 3, 2, 5)$ and $Q(0, 1, 1, 4)$ at $(4, 1, 2, 2, 1, 1)$ and $(2, 3, 2, 2, 1, 1)$. Reflection and interchange within a doubled pair cover every labelled placement, and fixed-parity monotonicity covers every further length increase. For all-odd K_4, the two-long paths are either adjacent or opposite. A nonconstant physical switch makes exactly three or four paths even, the edges of a nontrivial K_4 cut, and every even path is long. The regular simplex covers the physically all-odd $j \ge 3$ rows and every switched row with $j \ge 4$. The only remaining DNN row is a three-edge even cut and the complementary odd unit triangle. Relabel the cut as the star from vertex 0 and use

$$R = \begin{pmatrix} 1 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1 & -11/24 & -11/24 \\ 1/6 & -11/24 & 1 & -11/24 \\ 1/6 & -11/24 & -11/24 & 1 \end{pmatrix}.$$

Its eigenvalues are $35/24, 35/24, 13/12, 0$. At the worst physical lengths, the three even cut paths

have length two and the three triangle paths are odd units, giving

$$3\frac{38-8\sqrt{21}}{5}+3\frac{13}{35}=\frac{837-168\sqrt{21}}{35}<2,$$

because $168^2\cdot 21-767^2=4415>0$; fixed-physical-parity monotonicity covers all longer descendants. Hence the $j=0,1$ rows, and only those all-odd rows, require structural ownership. This also verifies directly that an external triangle closes by DNN every newly listed noncanonical state: each of

$$\frac{229}{120},\quad \frac{31}{20},\quad \frac{1862}{1000},\quad \frac{1662}{1000},\quad \frac{71}{40},\quad 24-16\sqrt{2},\quad \frac{837-168\sqrt{21}}{35}$$

is strictly below two, and $\epsilon_3=1$.

For a hostile cycle $Q=C_q$, where $q\equiv 1 \pmod 4$ and $q\geq 5$,

$$\epsilon_q\leq 5-2\sqrt{5}<\frac{3}{5}.$$

All DNN rows in the table have $e_B\leq 2$, so $e_B+\epsilon_q<3$. For the three non-K_4 structural types and the one-long all-odd K_4 type, the displayed bounds give respectively $221/100+3/5<3$, $9/4+3/5<3$, and $12/5+3/5=3$ with strictness in both constituent estimates. The attached K_4 packet has credit >2; an actual bridge separating it from Q is cut and its whole connector is assigned to one endpoint territory, giving total credit $>2-(\sec(\pi/q)-1)>0$. At a shared cut z, the K_4 territory owns z and every branch rooted on its side, while the other territory owns $Q-z$ and its branches; the latter is one nonempty tree, so the total is $>2-1>0$.

It remains to specify ownership for the structural rows when Q is a triangle or has length 3 (mod 4). For a canonical doubled-triangle or doubled-C_4 row, open exactly the internal path vertex used in its tricyclic repair; for a one-long all-odd K_4 row, open an internal vertex of the long path. Give the opened vertex every component whose unique block-cut route first meets the rank-three block there. The complementary territory contains the same attached two-triangle or diamond packet as in the tricyclic proof. If the opened territory owns Q, it is a favorable unicyclic packet of credit >0 and the complement has credit >1. Otherwise it is one nonempty tree of credit -1, while the complement has three favorable cycles. Packing number at most two gives complement credit >2 by the Sachs argument above; packing number three supplies an actual bridge separating credits >1 and >0. Every connector remnant stays whole on one side, every cut has one owner, and each rooted branch follows the territory owning its root. For $j=0$, the attached-K_4 territory owns the complete K_4 and its rooted branches; a separating route leaves a favorable unicyclic territory, while at a shared cut z the other territory is the tree $Q-z$. Even cycles have zero DNN excess. This exhausts the table and assigns every vertex, connector, cut, and branch exactly once; the underlying Gram records are reproduced in [15]. This proves every multiblock alternative. □

4 Classification of the rank-four kernels

It remains to consider one positive-rank block. Suppress every maximal path whose internal vertices have block degree two, retaining parallel kernel edges. The original graph is simple, so at most one path in a parallel class can have length one.

Proposition 4.1 (Seventeen-kernel classification)**.** *Let K be a loopless multigraph with at least two vertices, no cut vertex, minimum degree at least three, and cyclomatic rank four. Then $2\leq |V(K)|\leq 6$, and K is isomorphic to exactly one of the seventeen multigraphs in Table 1.*

Proof. Let $v = |V(K)|$ and $e = |E(K)|$. Since $e = v + 3$,

$$\sum_x (d(x) - 2) = 2e - 2v = 6. \tag{11}$$

Every summand is positive, so $2 \le v \le 6$. The positive integers $d(x) - 2$ partition six. For each resulting degree multiset introduce a nonnegative integer a_{ij} for every pair $i < j$ and solve

$$\sum_{j \neq i} a_{\min(i,j),\max(i,j)} = d_i.$$

Reject a solution if its positive support disconnects after deletion of some vertex, and quotient by permutations preserving the degree multiset. The possible degree rows and surviving orbit counts are

v	degree multisets	orbits
2	$(5,5)$	1
3	$(5,4,3), (4,4,4)$	$1+1$
4	$(5,3,3,3), (4,4,3,3)$	$1+4$
5	$(4,3,3,3,3)$	4
6	$(3,3,3,3,3,3)$	5.

Solving these incidence equations gives exactly the vectors in Table 1. Their lexicographically least vectors over all labellings are distinct, proving nonisomorphism. Suppression cannot create a loop for the same cut-vertex reason used in Proposition 3.1. The independently regenerated list has canonical checksum

`d89e6e60c66e480ba89e662ab90b5ace211cbcff7292f92ad1614bb0937eb8e9.`

□

5 Exact certificate ledgers for all seventeen kernels

A physical row records, in each kernel bundle, the number of odd paths. It is not a switching class. A certificate consists of an explicit correlation Gram matrix on branch vertices and, where needed, exact rational planar vectors on path interiors. Its cost is the sum in (5). Every accepted cost is at most the tetracyclic budget three. A canonical row uses length two for an even path and length one for an odd path, except that parallel odd paths use lengths one and three to preserve simplicity. A one-coordinate frontier adds two to one physical path. By Lemma 2.2, the canonical certificate covers a row unchanged in all coordinates, while any noncanonical descendant is covered by a frontier in one changed coordinate and monotonicity in every other coordinate.

Proposition 5.1 (Kernel certificate theorem)**.** *Every simple subdivision H of every kernel in Table 1 satisfies*

$$\kappa(H) \le |E(H)| + 3$$

or has an induced structural packet proving directly that $s^+(H) \ge |V(H)|$. The same conclusion holds after arbitrary rooted trees are attached at arbitrary vertices.

Proof. We summarize the exact ledgers by branch-vertex count. Each cited proof note prints the formulas or defines the digest-locked fixture and a fail-closed exact verifier.

Two branch vertices: kernel 1. The block is five internally disjoint terminal paths. Its excess is the five-term version of (7). Fixed-parity monotonicity reduces nonunit lengths to two or three. If no path is unit, endpoint tests cover four parity counts; in the remaining count the objective has negative derivative at the endpoint value three. If one path is unit, four parity counts are handled by $\theta = \pi$ or $2\pi/3$. In the last count, with $u = \tan(\theta/4)$, the upper bound is

$$\frac{1}{4u^2} - \frac{1}{2} + \frac{33}{4}u^2, \qquad \min = \frac{\sqrt{33}-1}{2} < 3.$$

Thus the excess is strictly below three for every simple five-path block [7].

Three branch vertices: kernels 2–3. Their bundle multiplicities are $(1, 2, 3)$ and $(2, 2, 2)$. If q of the six paths are odd and $q \geq 3$, the equicorrelation matrix with off-diagonal $-1/2$ gives cost

$$\frac{q}{3} + \frac{2(6-q)}{3} = 4 - \frac{q}{3} \leq 3.$$

For $q \leq 2$, the complete rational ledgers have respectively nine labelled bundle-count rows and four rows up to bundle permutation. All thirteen costs are strict; their maxima are $21/8$ and $17/8$. Every 3×3 Gram determinant is listed as a nonnegative rational number [8].

Four branch vertices: kernels 4–8. The physical row counts are

$$72,\ 72,\ 54,\ 48,\ 96, \qquad \text{total } 342.$$

The exact disjoint ledger is

$$342 = 270 \text{ base} + 70 \text{ rational patch} + 2 \text{ independently discharged.} \tag{12}$$

The last two rows are kernel 8 with all five singleton bundles odd and the doubled bundle containing one or two odd paths. An exhaustive $2 \cdot 2^6 = 128$ long/unit state census has 56 stabilizer orbits. There are 114 states with at least two long support paths, twelve one-long states in six planar Gram classes, and two no-long structural states. A regular-simplex matrix proves the first group strictly below three; exact planar angles with denominators $3, 5, 7$ prove the second; in the last group, deleting the extra path leaves an attached all-odd K_4 packet against one tree. The verifier proves all trigonometric comparisons with rational intervals and rejects eight hostile mutations. The canonical payload digest is

`f381b2b28bd3f45d7c96d90bce824a308bc340c9d6ebd098e5da116b89648d5a.`

See [9].

Five branch vertices: kernels 9–12. The physical and genuine automorphism-orbit ledgers are

kernel	9	10	11	12	total
physical rows	108	192	144	256	700
orbits	63	120	144	51	378.

An exact equilateral three-color sieve, checking all $3^5 = 243$ labelled colorings per orbit, gives

$$378 = 370 \text{ direct} + 8 \text{ residual.} \tag{13}$$

The physical odd–odd doubled-bundle cost uses lengths $(1, 3)$ and the algebraic number $a = 3\tan^2(\pi/18)$, isolated as the unique root in $(93/1000, 94/1000)$ of $a^3 - 27a^2 + 99a - 9$. Thus no floating comparison occurs.

The eight residual orbits split as $4, 2, 2, 0$ across kernels 9–12. Seven non-symbolic rows use sixty exact rational records: canonical and one-coordinate $+2$ frontiers for six eight-path rows, plus six doubled-path frontiers for one kernel-9 row. The remaining kernel-9 equality row uses five planar vectors over $\mathbb{Q}(\sqrt{3})$: two odd singleton paths have correlation -1 and cost zero, while each of three mixed doubled bundles has correlation $-1/2$ and cost $1/3 + 2/3 = 1$. Its total is exactly three. The closure verifier recomputes every rational internal-vector ledger and the symbolic Gram identities and rejects ten hostile mutations [10, 11].

Six branch vertices: kernels 13–17. Kernel 16 is $K_{3,3}$. Its $2^9 = 512$ physical parity rows have exact three-color score histogram

S	0	3	4	5	6	7	8	9
#	1	6	27	63	147	168	81	19.

If q odd paths and b even paths are bichromatic, the excess is $(q + 2b)/3 = S/3 \le 3$. The verifier checks all $3^6 = 729$ colorings for every row and all 26 genuine automorphism orbits [12].

For kernels 13, 14, 15, and 17, the labelled physical-row counts are $288, 216, 384, 512$, the automorphism-group orders are $4, 6, 4, 12$, and the genuine orbit counts are $102, 56, 144, 74$. Thus there are 1400 labelled rows and 376 genuine orbits. The exact partition is

$$376 = 359 \text{ three-color certified} + 17 \text{ residual}, \qquad (5, 6, 5, 1) \text{ residual by kernel.} \tag{14}$$

For the sixteen residual orbits of kernels 13–15, canonical and nine one-coordinate frontiers give

$$160 = 148 \text{ strict rational certificates} + 12 \text{ symbolic kernel-14 targets.} \tag{15}$$

The twelve targets are the canonical row and coordinates $0, 3, 8$ for each of three kernel-14 rows. Their three Gram determinants are $1/2, 0, 1/2$; every 2×2 principal minor is $3/4$, each mixed doubled bundle costs one, and each canonical total is exactly three.

Kernel 17 has 512 labelled parity rows and 74 genuine automorphism orbits. Seven exact planar templates over $\mathbb{Q}(\sqrt{3})$, with all 32 branch sign switches per template, cover all 512 physical rows. Their disjoint first-cover gains are

$$284,\ 123,\ 56,\ 24,\ 13,\ 8,\ 4,$$

which sum to 512. In particular, they cover the residual orbit at its canonical vector and all nine coordinate frontiers. The final verifier requires the 359-row sieve, 148-record fixture, twelve symbolic targets, and seven-template audit, then reconstructs all ledgers. The residual is zero [13].

All direct rows now satisfy (6). Every structural row uses an induced partition whose deleted territory and all branches based there are owned explicitly; retained branches stay with the complementary packet. Fixed-parity monotonicity proves arbitrary subdivisions, and Lemma 2.1 proves arbitrary rooted-tree attachments. This completes all seventeen kernels. □

Proof of Theorem 1.1. Apply Proposition 3.1. If there is more than one positive-rank cyclic block, Proposition 3.2 applies. If there is one rank-four block, suppress its degree-two paths. Proposition 4.1 gives one of the seventeen kernels, and Proposition 5.1 proves the required inequality for every subdivision and every attached rooted tree. The alternatives are exhaustive, so $s^+(G) \ge |V(G)|$. □

A Verifier manifest and exact scopes

All commands below are run from the repository root. Run every finite audit normally and with Python assertions disabled; the paired outputs must be byte-identical where the verifier says so. The first four commands audit the new tricyclic theorem used in Lemma 1.3 and in the rank-$3+1$ multiblock rows; the remaining commands audit the rank-four kernel proof.

```
python3 research/tricyclic-finite-rational-certificates-verifier.py
python3 -O research/tricyclic-finite-rational-certificates-verifier.py
python3 positive-square-energy/experiments/k4_all_odd_exact_verify.py
python3 -O positive-square-energy/experiments/k4_all_odd_exact_verify.py

python3 research/rank-four-kernel-census-verifier.py
python3 -O research/rank-four-kernel-census-verifier.py

python3 research/rank-four-three-vertex-tables-verifier.py
python3 -O research/rank-four-three-vertex-tables-verifier.py

python3 research/rank-four-four-vertex-theorem-verifier.py
python3 -O research/rank-four-four-vertex-theorem-verifier.py

python3 research/rank-four-five-vertex-three-color-verifier.py
python3 -O research/rank-four-five-vertex-three-color-verifier.py
python3 research/rank-four-five-vertex-residual-closure-verifier.py
python3 -O research/rank-four-five-vertex-residual-closure-verifier.py

python3 research/rank-four-kernel16-three-color-verifier.py
python3 -O research/rank-four-kernel16-three-color-verifier.py
python3 research/rank-four-cubic-kernels-final-verifier.py
python3 -O research/rank-four-cubic-kernels-final-verifier.py

python3 research/tetracyclic-master-verifier.py
python3 -O research/tetracyclic-master-verifier.py
```

The tricyclic audits reconstruct the finite physical-row ledgers used by the companion theorem. Their expected canonical payload digests are, respectively,

```
6d499af761ffa7317a431e5150dac59317618f508896be2c3ed77b250b43c901
b0c4ed64a12ccabe0a8a38cb9fdb85622d8f7c2d07bdf53f75eb1700a684fa39.
```

They are dependencies of the multiblock proof, not part of the seven-verifier rank-four master manifest. The census verifier independently enumerates weak edge-multiplicity compositions on two through six vertices. It finds respectively $1, 7, 54, 255, 550$ labelled survivors and collapses them to $1, 2, 5, 4, 5$ canonical classes. It checks the exact list, degree excess, simplicity costs, checksum, and hostile mutations.

The three-vertex verifier reconstructs all $24 + 27 = 51$ physical rows, split as $9 + 10$ low-odd rational-table rows and $15 + 17$ common-simplex rows, and checks every Gram principal minor exactly. The four-vertex verifier reconstructs the disjoint $342 = 270 + 70 + 2$ ledger and all 128 long/unit frontier states. The five-vertex pair reconstructs all 700 physical rows, 378 orbits, the

$370 + 8$ sieve, the sixty rational records, and the symbolic equality row. The kernel-16 verifier reconstructs all 512 rows and its score histogram. The final cubic verifier invokes its dependency audits and checks $376 = 359 + 17$, $160 = 148 + 12$, all symbolic kernel-14 matrices, and all 512 kernel-17 first-cover assignments. The master verifier invokes all seven finite dependency verifiers, verifies their file and output digests, checks that kernels 1 through 17 have exactly one theorem owner (with the stated sieve/closure composition for kernels 9–12), and records kernel 1 as the separate analytic five-path theorem.

These programs are exact audits of finite combinatorial and rational or quadratic-algebraic proof objects. They do not replace the mathematical proofs of DNN duality, path elimination, monotonicity, block classification, induced superadditivity, or the multiblock territory lemmas. Conversely, the paper does not ask a floating-point optimizer or finite graph-order census to prove an all-length assertion.

The exact rank-four dependency manifest printed by the master verifier has SHA-256 digest

`9fa7bdf4a4a296a69f818bf78d5fe1a3aba5bddb38639ea784593d0291dfe19f.`

The kernel-census payload itself retains the checksum printed in Proposition 4.1; the manifest digest changed when that verifier gained automorphism and minimum-simple-subdivision checks and the finite dependencies gained stricter exact-type checks. The master now pins seven direct verifier outputs, eleven transitive dependency files, and three nested verifier outputs; it also records the historical direct manifest.

The detailed proof-object sources are:

```
positive-square-energy/tetracyclic-general/
  five-path-dnn-theorem.md
  three-vertex-kernels-dnn.md
  four-vertex-kernels-dnn.md
  five-vertex-three-color-dnn-sieve.md
  five-vertex-residual-closure-theorem.md
  kernel16-three-color-dnn.md
  cubic-kernels-13-15-17-final-theorem.md
  rank-four-kernel-classification.md
  theta-two-cycle-residuals.md
  two-diamond-blocks.md
  rank3-hostile-cycle-packet.md
```

Build this manuscript with

```
bash scripts/build-paper.sh all-tetracyclic-graphs
```

Scoped nonclaim. The theorem concerns exactly finite simple connected graphs with $m = n+3$. The argument does not assume that adding an edge increases s^+, does not transport physical path lengths through switching, and does not infer a quantitative reserve from an unquantified lower-rank theorem. Several DNN certificates attain their auxiliary excess budget three. We therefore assert only $s^+(G) \geq n$ globally, not strict inequality and not a classification of spectral equality.

AI disclosure

This manuscript and its proof-object synthesis were generated by an AI coding and mathematical reasoning system from the cited repository materials. No human authorship, human review, independent verification, or conventional peer review is claimed. The exact scripts are corroborative

only; the result should be assessed from the displayed argument and its stated local dependencies. The repository post and result card are announcements of this AI-generated manuscript, not evidence of external validation or publication.

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[3] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.

[4] Companion manuscript, *Positive Square Energy of Every Bicyclic Graph*, 2026; `all-bicyclic-graphs/paper.tex`.

[5] Companion manuscript, *Positive Square Energy of Every Tricyclic Graph*, 2026; `all-tricyclic-graphs/paper.tex`.

[6] Companion manuscript, *Positive Square Energy of Every Tetracyclic Cactus*, 2026; `all-tetracyclic-cacti/paper.tex`.

[7] *The five-path rank-four block*, proof note, 2026; `positive-square-energy/tetracyclic-general/five-path-dnn-theorem.md`.

[8] *Three-vertex rank-four kernels: a uniform DNN certificate*, proof note, 2026; `positive-square-energy/tetracyclic-general/three-vertex-kernels-dnn.md`.

[9] *Four-vertex rank-four kernels: complete exact ledger*, proof note, 2026; `positive-square-energy/tetracyclic-general/four-vertex-kernels-dnn.md`.

[10] *Five-vertex rank-four kernels: exact coarse three-color sieve*, proof note, 2026; `positive-square-energy/tetracyclic-general/five-vertex-three-color-dnn-sieve.md`.

[11] *Five-vertex rank-four kernels: closure of the eight residual rows*, proof note, 2026; `positive-square-energy/tetracyclic-general/five-vertex-residual-closure-theorem.md`.

[12] *Kernel 16: exact three-color certificate on all physical rows*, proof note, 2026; `positive-square-energy/tetracyclic-general/kernel16-three-color-dnn.md`.

[13] *Cubic kernels 13–15 and 17: exact final theorem*, proof note, 2026; `positive-square-energy/tetracyclic-general/cubic-kernels-13-15-17-final-theorem.md`.

[14] *The tetracyclic* $2 + 1 + 1$ *residuals*, proof note, 2026; `positive-square-energy/tetracyclic-general/theta-two-cycle-residuals.md`.

[15] *Rank-three block plus one hostile cycle: canonical residual packet*, proof note, 2026; `positive-square-energy/tetracyclic-general/rank3-hostile-cycle-packet.md`.

[16] *Two diamond blocks*, proof note, 2026; `positive-square-energy/tetracyclic-general/two-diamond-blocks.md`.

No.	v	degree multiset	upper-triangle multiplicities	simplicity cost
1	2	$5,5$	5	4
2	3	$5,4,3$	$1,2,3$	3
3	3	$4,4,4$	$2,2,2$	3
4	4	$5,3,3,3$	$0,1,2,1,2,1$	2
5	4	$4,4,3,3$	$0,1,2,2,1,1$	2
6	4	$4,4,3,3$	$0,1,2,2,2,0$	3
7	4	$4,4,3,3$	$0,1,2,3,1,0$	3
8	4	$4,4,3,3$	$1,1,1,1,1,2$	1
9	5	$4,3,3,3,3$	$0,0,1,2,1,0,2,2,0,0$	3
10	5	$4,3,3,3,3$	$0,0,1,2,1,1,1,1,1,0$	1
11	5	$4,3,3,3,3$	$0,0,1,2,1,1,1,2,0,0$	2
12	5	$4,3,3,3,3$	$0,1,1,1,1,1,1,0,1,1$	0
13	6	$3,3,3,3,3,3$	$0,0,0,1,2,0,1,1,1,2,1,0,0,0,0$	2
14	6	$3,3,3,3,3,3$	$0,0,0,1,2,0,1,2,0,2,0,1,0,0,0$	3
15	6	$3,3,3,3,3,3$	$0,0,0,1,2,1,1,0,1,1,1,0,1,0,0$	1
16	6	$3,3,3,3,3,3$	$0,0,1,1,1,0,1,1,1,1,1,1,0,0,0$	0
17	6	$3,3,3,3,3,3$	$0,0,1,1,1,1,0,1,1,1,0,1,1,0,0$	0

Table 1: Canonical kernel list. Multiplicities use lexicographic upper-triangle order; the final column is $\sum_{u<v} \max(m_{uv} - 1, 0)$.